Tel Aviv University | Faculty of Engineering

Linear Algebra for Electrical Engineering

Course 05091724 | Week 1

Sets, Logic, Induction and Fields

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Notes

- This document contains a preliminary, non-binding outline of the material to be covered in the lectures.
- There may be omissions, errors, and other differences between these notes and the material actually taught.
- Proofs will be presented in the lectures. You are expected to know and understand them. Further details and examples will also be given in class.
- The material actually covered in the lectures is the authoritative course material.
- Reports of errors are very helpful and are welcome at yohaim@tauex.tau.ac.il.

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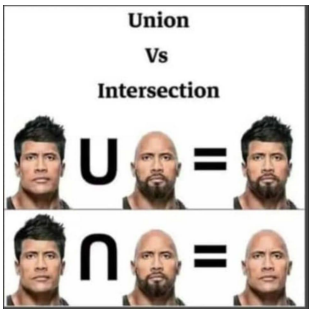
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1 Set Theory, Logic, Proofs, Induction and Newton's Binomial Theorem

1.1 Set Theory

Sets

A set in mathematics may be a collection of numbers, such as

$$A = \{1, 2, 6\},$$

or a collection of more complicated objects. Almost any mathematical object can belong to a set, or be an *element* of a set.

There are several ways to describe a set. We can list its elements, sometimes using an implied pattern, especially when the set is infinite. We can describe its elements in words. We can also describe it as a subset of another set by specifying a condition for membership.

Examples.

1. The set $\{1, 2, 6\}$ above is described by listing all its elements. It has three elements. For example, the number 4 does not belong to it, whereas 2 does.

2. The set $\{1, 2, \ldots, 10\}$ is described by an implied pattern. The number 7 belongs to it, whereas 11 does not. It contains ten numbers.

3. Avoid an implied pattern when it could lead to confusion about membership. For example, does 6 belong to $\{2, 4, \ldots\}$?

   If this is the set of positive even integers, then it does. But the intended set might instead be the powers of 2,

   $$\{2, 4, 8, 16, \ldots\},$$

   in which case it does not. A verbal description, such as "the set of all positive even integers," is better in this situation.

4. The set of colors of the Israeli flag is $\{\text{white}, \text{blue}\}$.

5. The empty set, denoted by $\varnothing$, is also a set. It is simply empty: it has no elements.

6. The elements of a set can themselves be sets. For example,

   $$C = \big\{\{1, 2, 6\}, \{1, 3\}\big\}$$

   is a set with two elements.

   It is easy to get confused here. Does the set $\{2, 6\}$ belong to C? No. It is not one of the listed elements. It is contained in one of the elements, a relation we will discuss shortly.

7. A set can also be described by a membership condition. For example, if we temporarily agree that our universe consists only of numbers on the real line, then

   $$A = \{x \mid x > 0\}.$$

   The vertical bar $\mid$ is read "such that." Thus this is the set of all positive numbers.

It is important to remember that a set has no repetitions and that order does not matter. All the following sets are exactly the same:

$$\begin{aligned}\{1, 2, 6\} &= \{2, 1, 6\} = \{1, 2, 1, 6\} \\ &= \{6, 6, 2, 1, 2, 1, 6, 1, 2, 1, 6\}.\end{aligned}$$

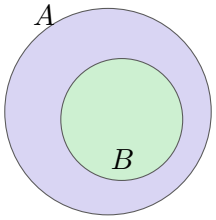
Definition 1. We write $a \in A$ when a belongs to the set A, or equivalently, when a is an element of A. Otherwise, we write $a \notin A$.

Example. For instance,

$$1 \in \{1, 2, 6\}, \qquad 10 \notin \{1, 2, 6\}.$$

Subsets

Definition 2. Let A and B be sets. We say that B is *contained in* A, or that B is a *subset* of A, and write $B \subseteq A$, if every element of B is also an element of A. In other words, for every $b \in B$, we have $b \in A$.



Remark. The Hebrew notes use both $B \subset A$ and $B \subseteq A$ for inclusion allowing equality. In these English notes, we use $\subseteq$ consistently. In particular, $B = A$ is allowed.

Examples.

1. We have $\{1, 2\} \subseteq \{1, 2, 6\}$. Since these sets are not equal, one sometimes writes

   $$\{1, 2\} \subsetneq \{1, 2, 6\}.$$

   We will usually use the weaker inclusion notation, which permits equality.

2. The set $\{1, 3\}$ is not contained in $\{1, 2, 6\}$. We can write

   $$\{1, 3\} \not\subseteq \{1, 2, 6\}.$$

3. Every set A satisfies $A \subseteq A$.

4. If $A = \{1, 2, 6\}$, we can define a subset of A by a condition. For example,

   $$B = \{a \in A \mid a \text{ is even}\} = \{2, 6\}.$$

   Again, the vertical bar means "such that."

5. The set

   $$F = \big\{\{1, 2, 6\}, \{2, 6\}\big\}$$

   has sets as its elements. In the notation of the preceding example, both A and B belong to F. Does $\{1, 2\}$ belong to F?

6. Let $A = \{1, 2\}$ and $B = \big\{\{1, 2\}, \{1, 3\}\big\}$. Then

   $$A \in B, \qquad A \not\subseteq B.$$

Remark. Other sources sometimes use $\subset$ and $\subsetneq$ with the same meaning. Always check the convention being used.

The Power Set

Definition 3. Let A be a set. The *power set* of A is the set of all subsets of A. It is denoted by $\mathcal{P}(A)$.

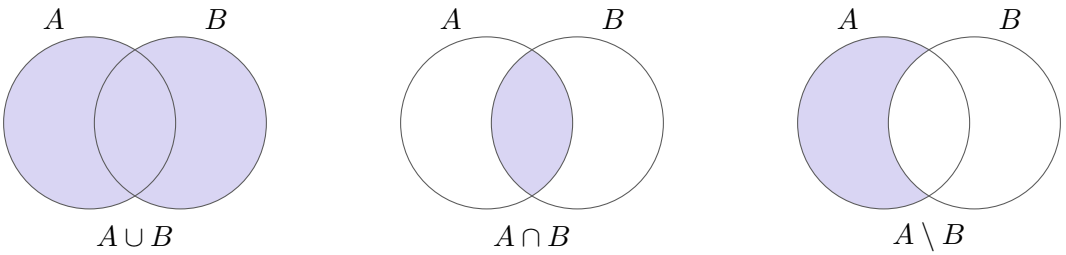
Example. For $A = \{1, 2, 6\}$,

$$\mathcal{P}(A) = \big\{\varnothing, \{1\}, \{2\}, \{6\}, \{1, 2\}, \{1, 6\}, \{2, 6\}, \{1, 2, 6\}\big\}.$$

Operations on Sets

Definition 4. Let A and B be sets.

1. The *union* of A and B is
$$A \cup B = \{c \mid c \in A \text{ or } c \in B\}.$$
2. The *intersection* of A and B is
$$A \cap B = \{c \mid c \in A \text{ and } c \in B\}.$$
3. The *difference* of A and B is
$$A \setminus B = \{c \mid c \in A \text{ and } c \notin B\}.$$



Examples.

1. Let $A = \{1, 2, 6\}$ and $B = \{1, 3\}$. Then
$$A \cup B = \{1, 2, 3, 6\}, \quad A \cap B = \{1\},$$
$$A \setminus B = \{2, 6\}, \qquad B \setminus A = \{3\}.$$
A set with exactly one element is called a *singleton*.
2. Let
$$A = \big\{\{1, 2, 6\}, \{1, 7\}\big\}, \qquad B = \big\{\{1, 3\}, \{7\}\big\}.$$
What is $A \cap B$?

   The elements of these sets are themselves sets, so the question is: which sets occur as elements of both A and B? None do. Therefore $A \cap B = \varnothing$. In this case, A and B are called *disjoint*.

   It is tempting to think that $\{1\}$, for example, belongs to the intersection. But it is not an element of either A or B. Their elements are sets; we are not intersecting the elements themselves.

Sometimes we agree to work only with subsets of a particular, clearly specified set A. In that setting, the *complement* of a set $B \subseteq A$ is the difference $A \setminus B$. We denote the complement of B by B^c.

Example. In first-grade arithmetic, the only numbers considered are the nonnegative integers. Thus every set under discussion, such as the even numbers or the odd numbers, is a subset of

$$A = \{0, 1, 2, \ldots\}.$$

In this context, for example,

$$\{1, 2, 3, \ldots\}^c = \{0\}.$$

Remark. Another common notation for the complement of B is $\overline{B}$. We will not use it here. If the larger set A is not completely clear, it is better to write $A \setminus B$ explicitly rather than speak of a complement.

Sets of Numbers

The real numbers are more elusive than the simpler number systems we will list below. Everyone knows π, an excellent example of a real number. But what exactly is meant by "the real numbers"? To see what the question is asking, consider some important number systems in historical order.

Name	Symbol	Description	Comments
Natural numbers	$\mathbb{N}$	The numbers $1, 2, 3, \ldots$	The most basic numbers, used for counting.
Integers	$\mathbb{Z}$	The natural numbers, together with 0 and the negative integers: $\ldots, -2, -1, 0, 1, 2, \ldots$	These numbers arise naturally, for example when keeping track of debts.
Rational numbers	$\mathbb{Q}$	Ratios m/n of integers, where $m, n \in \mathbb{Z}$ and $n \neq 0$	These also arise naturally: half an apple, for example, or solutions of equations $nx = m$ with integer coefficients.
Real numbers	$\mathbb{R}$	All numbers on the real line	This is not a precise construction, but it will suffice here.
Complex numbers	$\mathbb{C}$	Numbers $a + bi$, where $a, b \in \mathbb{R}$ and $i^2 = -1$	Further details will be given in the tutorial.

Remarks.

- In our convention, 0 is not a natural number: $0 \notin \mathbb{N}$. Some sources include it, so there is no universal convention.
- The rational expression m/n is not defined when $n = 0$. It is enough to allow the numerator m to be negative in order to represent all possible ratios of two integers. For example,

  $$\frac{2}{-4} = -\frac{1}{2}.$$

  We therefore usually choose a *canonical representation*, meaning one that does not depend on an arbitrary choice: $m \in \mathbb{Z}$, $n \in \mathbb{N}$, and m, n relatively prime. Every rational number has a unique canonical representation of this kind.
- Could $\mathbb{Q}$ contain every number on the real line? For example, could π be a ratio of two integers? In other words, could $\mathbb{R} = \mathbb{Q}$?

  It turns out that it does not. The numbers $\sqrt{2}$, π, and e are examples of real numbers that are not rational. A real number that is not rational is called *irrational*.

Equality of Sets

Given two sets A and B, the usual way to prove that they are equal is to prove inclusion in both directions:

$$A \subseteq B \qquad \text{and} \qquad B \subseteq A.$$

Example. Let $A = \mathbb{Q}$, and let B be the set of real numbers x that solve a nontrivial linear equation with integer coefficients,

$$ax + b = 0, \qquad a, b \in \mathbb{Z}.$$

More precisely,

$$B = \{x \in \mathbb{R} \mid \exists a, b \in \mathbb{Z},\ (a, b) \neq (0, 0),\ ax + b = 0\}.$$

We will prove that $A = B$ by proving both inclusions.

Proof. First we prove $A \subseteq B$. Let $x \in A$. Since $A = \mathbb{Q}$, the number x is rational. Thus there are integers m, n, with $n \neq 0$, such that

$$x = \frac{m}{n}.$$

It follows that $m = nx$, and hence $nx - m = 0$. Set $a = n$ and $b = -m$. Then a, b are integers, they are not both zero, and $ax + b = 0$. Therefore $x \in B$. This proves $A \subseteq B$.

Now we prove $B \subseteq A$. Let $x \in B$. By definition, there are integers a, b, not both zero, for which $ax + b = 0$.

If $a = 0$, this equation gives $b = 0$. That would mean $a = b = 0$, which is impossible. Therefore $a \neq 0$, and we may write

$$x = -\frac{b}{a}.$$

Since a and $-b$ are integers, this ratio belongs to $\mathbb{Q}$. Hence $x \in A$, proving $B \subseteq A$.

We have established both inclusions, so $A = B$. □

1.2 Logic

Quantifiers and Formulas

The symbols $\forall$ and $\exists$ are called *quantifiers*.

- The symbol $\forall$ means "for every." For example, $\forall n \in \mathbb{N}$ means "for every natural number n."
- The symbol $\exists$ means "there exists." For example, $\exists n \in \mathbb{N}$ means "there exists a natural number n."

These quantifiers allow us to form mathematical formulas.

Examples.

1. The statement $\forall x \in \mathbb{R} :\ x^2 \geq 0$ is true.
2. The corresponding expression $\forall x \in \mathbb{C} :\ x^2 \geq 0$ cannot be used as a general ordered inequality: $i^2 = -1$. Here inequalities refer to the usual order on $\mathbb{R}$; $\mathbb{C}$ is not an ordered field.

The symbol $\Rightarrow$, or sometimes $\rightarrow$, means "implies." We will usually use $\Rightarrow$.

Examples.

1. Consider the formula
$$\forall x \in \mathbb{R}: \quad x < 2 \;\Rightarrow\; x^2 < 2.$$
Is it true? No: take, for example, $x = -5$.

2. Here are several similar formulas.

Formula	Truth value
$\forall x \in \mathbb{R}\ \forall n \in \mathbb{Z}:\ x + n > 0$	False
$\forall n \in \mathbb{Z}\ \forall x \in \mathbb{R}:\ x + n > 0$	False
$\forall x \in \mathbb{R}\ \exists n \in \mathbb{Z}:\ x + n > 0$	True
$\exists n \in \mathbb{Z}\ \forall x \in \mathbb{R}:\ x + n > 0$	False
$\forall n \in \mathbb{Z}\ \exists x \in \mathbb{R}:\ x + n > 0$	True
$\exists x \in \mathbb{R}\ \forall n \in \mathbb{Z}:\ x + n > 0$	False

Two universal quantifiers may be interchanged without changing the meaning, as may two existential quantifiers. The resulting formulas are equivalent. In general, a universal and an existential quantifier cannot be interchanged.

And, Or

The symbol $\wedge$ means "and," and $\vee$ means inclusive "or." These symbols are called *logical connectives*.

Example. The formula
$$\forall x \in \mathbb{R}\ \exists n \in \mathbb{N}: \quad \lfloor x \rfloor = n \ \wedge\ n \le x$$
says that for every real number x, there is a natural number n equal to the integer part of x and also less than or equal to x. Here $\lfloor x \rfloor$ denotes the greatest integer not exceeding x, written $[x]$ in the Hebrew notes.

This formula is false. For example, if $x = -1.3$, then $\lfloor x \rfloor = -2$, so there is no natural number n equal to $\lfloor x \rfloor$.

More precisely, each statement has a truth value: true or false. The connectives $\wedge$ and $\vee$ are defined by their truth tables. We use T for true and F for false.

P	Q	$P \wedge Q$	$P \vee Q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Negation

The logical connective $\neg$ means negation.

P	$\neg P$
T	F
F	T

Examples.

1. The fragment $\neg(\exists n \in \mathbb{N})$ is read "there does not exist a natural number n." No property has been specified, so this is not a complete statement whose truth we can assess.

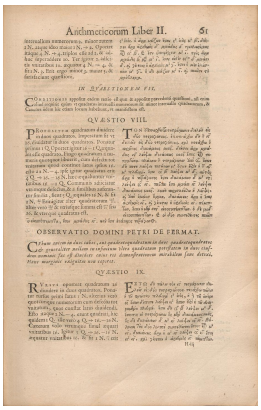
2. The statement
$$\forall n \in \mathbb{N}: \quad n > 2 \;\Rightarrow\; \neg\big(\exists x, y, z \in \mathbb{N}: \; x^n + y^n = z^n\big)$$
is known as *Fermat's Last Theorem*.

   The notation $\exists x, y, z \in \mathbb{N}$ abbreviates
$$\exists x \in \mathbb{N}\ \exists y \in \mathbb{N}\ \exists z \in \mathbb{N}.$$
   As we have seen, the order of these three existential quantifiers does not matter. They say that there exist natural numbers x, y, z.

   What does the statement mean? If $n > 2$ is a natural number, there are no natural numbers x, y, z satisfying $x^n + y^n = z^n$.

   For comparison, when $n = 2$ we have the Pythagorean equation,
$$x^2 + y^2 = z^2.$$
   The legs x, y and hypotenuse z of a right triangle satisfy this equation. There are right triangles with integer side lengths; for example, $x = 3$, $y = 4$, $z = 5$. There are infinitely many such triples, called *Pythagorean triples*.

   Fermat conjectured that $x^3 + y^3 = z^3$ has no solution in natural numbers. In his copy of the famous book *Arithmetica*, he wrote a remark to the effect that he had a truly marvelous proof, but the margin was too narrow to contain it.


Arithmeticorum Liber II. 61

QVÆSTIO VIII.

OBSERVATIO DOMINI PETRI DE FERMAT.

QVÆSTIO IX.

   The same assertion holds for every integer $n \geq 3$. Mathematicians devoted enormous effort to finding a proof; Andrew Wiles finally completed a correct proof in 1994.

   Consider the inner expression
$$\neg\big(\exists x, y, z \in \mathbb{N}: \; x^n + y^n = z^n\big).$$
   We can express it differently. If no natural numbers x, y, z satisfy the equality, then every choice of natural numbers fails to satisfy it. Thus
$$\neg\big(\exists x, y, z \in \mathbb{N}: \; x^n + y^n = z^n\big) \;\equiv\; \forall x, y, z \in \mathbb{N}: \; x^n + y^n \neq z^n.$$
   The symbol $\equiv$ means "is equivalent to." Fermat's Last Theorem can therefore also be written as
$$\forall n \in \mathbb{N}: \quad n > 2 \;\Rightarrow\; \big(\forall x, y, z \in \mathbb{N}: \; x^n + y^n \neq z^n\big).$$

Equivalence of Formulas

The notation $P \equiv Q$ expresses that P and Q have the same truth value: both are true or both are false. More precisely, when P and Q are built from basic statements whose truth values are not fixed, we write $P \equiv Q$ if they have the same truth table for all possible truth values of those basic statements.

Rules for Negating a Formula

Negating a quantified formula interchanges $\forall$ and $\exists$. More precisely,

1. $\neg(\forall x\ P(x)) \equiv \exists x\ \neg P(x)$.
2. $\neg(\exists x\ P(x)) \equiv \forall x\ \neg P(x)$.
3. $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$.
4. $\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q)$.
5. $\neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q)$.

To understand the third rule, which is often the most challenging, consider the truth table for $P \Rightarrow Q$:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

The implication fails only when P is true and Q is false. In other words, its negation is $P \wedge \neg Q$.

Examples.

1. We saw that $\exists n \in \mathbb{Z}\ \forall x \in \mathbb{R}:\ x + n > 0$ is false. Its negation is therefore true:

$$\forall n \in \mathbb{Z}\ \exists x \in \mathbb{R}: \quad x + n \leq 0.$$

   Indeed, for every integer n there is a real number x for which $x + n$ is nonpositive.

2. Fermat's Last Theorem is

$$\forall n \in \mathbb{N}: \quad n > 2 \ \Rightarrow\ \big(\forall x, y, z \in \mathbb{N}:\ x^n + y^n \neq z^n\big).$$

   Its negation is

$$\exists n \in \mathbb{N}: \quad n > 2 \ \wedge\ \big(\exists x, y, z \in \mathbb{N}:\ x^n + y^n = z^n\big).$$

   This says that there is a natural number $n > 2$ for which the equation has a solution in natural numbers. As noted above, that statement is false.

3. The formula

$$P: \quad \forall x \in \mathbb{R}\ \exists n \in \mathbb{N}:\ \lfloor x \rfloor = n$$

   is false. Hence its negation,

$$\neg P: \quad \exists x \in \mathbb{R}\ \forall n \in \mathbb{N}:\ \lfloor x \rfloor \neq n,$$

   is true. It says that some real number x has integer part different from every natural number. For example, $x = -1.3$ has $\lfloor x \rfloor = -2$, and no natural number equals -2.

Proof by Contrapositive

A basic and important fact is that $P \Rightarrow Q$ is always equivalent to $\neg Q \Rightarrow \neg P$:

$$(P \Rightarrow Q) \equiv (\neg Q \Rightarrow \neg P).$$

This can be checked using truth tables:

P	Q	$\neg P$	$\neg Q$	$\neg Q \Rightarrow \neg P$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

The last column is the same as the truth table for $P \Rightarrow Q$. Therefore, proving $\neg Q \Rightarrow \neg P$ proves $P \Rightarrow Q$. The implication $\neg Q \Rightarrow \neg P$ is called the *contrapositive* of $P \Rightarrow Q$.

Examples.

1. Suppose, for the purpose of this logic example, that every dog has four legs. Let P mean "I am a dog" and Q mean "I have four legs." The assumption is $P \Rightarrow Q$.

   The contrapositive is $\neg Q \Rightarrow \neg P$: if I do not have four legs, then I am not a dog. This differs from the converse $Q \Rightarrow P$. Having four legs does not imply that I am a dog; I might, for example, be a cat.

2. Let n be a natural number and let p be prime, meaning a natural number greater than 1 whose only positive divisors are 1 and itself. We claim

   $$p \mid n^2 \;\Rightarrow\; p \mid n.$$

   For example, if n^2 is divisible by 3, then n is divisible by 3. A more familiar case is that if n^2 is even, then n is even.

   It may be easier to prove the equivalent contrapositive. Here we will prove

   $$p \nmid n \;\Rightarrow\; p \nmid n^2.$$

   We use a theorem that may be proved later: every natural number $n > 1$ can be written uniquely, up to reordering, as a product of primes. A prime divides n exactly when it occurs in that factorization.

   Assume $p \nmid n$. The case $n = 1$ is immediate. For $n > 1$, write

   $$n = q_1 \cdots q_k,$$

   where $q_1, \ldots, q_k$ are primes. By assumption, p is not among these primes. Therefore

   $$n^2 = q_1 q_1 \cdots q_k q_k$$

   is also a prime factorization without a factor p. Hence $p \nmid n^2$. We have proved the contrapositive, and therefore the original implication.

In general, proving the contrapositive is an equivalent way to prove an implication. It is not always the most convenient method, but sometimes it is.

Proof by Contradiction

Proof by contradiction is a powerful and useful method. It is sometimes confused with proof by contrapositive, although it is more general.

Suppose we want to prove $P \Rightarrow Q$. In a proof by contradiction, we assume $P \wedge \neg Q$, the negation of the desired implication. This is the *contradiction assumption*. If it leads to a false statement, called a *contradiction*, the assumption must be false, and the original implication is true.

Example. We will prove that if $r \in \mathbb{Q}$, then $r^2 \neq 2$. Equivalently, $\sqrt{2}$ is not rational.

According to an urban legend, probably untrue, a member of the Pythagorean school named Hippasus showed that a number in the "real world" need not be rational. Unfortunately for him, the story says that he did so on a boat, surrounded by enough Pythagoreans, perhaps even their master himself. He then found himself on his way to the bottom of the lake.

Proof. Suppose, for a contradiction, that a rational number r satisfies $r^2 = 2$. Write $r = m/n$ in canonical form. Then

$$\frac{m^2}{n^2} = 2, \qquad \text{so} \qquad m^2 = 2n^2.$$

Thus m^2 is divisible by 2. Since 2 is prime, m is also divisible by 2. Write $m = 2k$ for an integer k. Substitution gives

$$2n^2 = m^2 = (2k)^2 = 4k^2,$$

and hence $n^2 = 2k^2$. Therefore n^2 is even, so n is even as well.

Both m and n are even. This contradicts their being relatively prime in the canonical representation. Therefore no rational number satisfies $r^2 = 2$. □

1.3 Induction and Newton's Binomial Theorem

The Principle of Induction and Proof by Induction

The principle of mathematical induction looks simple and innocent. Suppose $A \subseteq \mathbb{N}$ satisfies:

1. $1 \in A$.
2. For every $n \in \mathbb{N}$, if $n \in A$, then $n + 1 \in A$.

Then $A = \mathbb{N}$.

This principle becomes particularly powerful as a method of proof. Suppose we have a sequence of statements $P(1), P(2), \ldots$. For example, $P(n)$ could be

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof by induction says that if

1. $P(1)$ is true, the *base case*; and
2. $P(n) \Rightarrow P(n+1)$ for every $n \in \mathbb{N}$, the *induction step*;

then $P(n)$ is true for every $n \in \mathbb{N}$.

There are therefore two stages in an induction proof. First prove $P(1)$. Then prove the induction step by assuming $P(n)$, the *induction hypothesis*, and using it to prove $P(n+1)$.

Examples.

1. We prove by induction that, for every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Base case. For $n = 1$,

$$1 = \frac{1 \cdot (1+1)}{2}.$$

Induction step. Fix $n \in \mathbb{N}$ and assume

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

We must prove

$$1 + 2 + \cdots + (n+1) = \frac{(n+1)(n+2)}{2}.$$

Using the induction hypothesis,

$$\begin{aligned} 1 + 2 + \cdots + (n+1) &= (1 + 2 + \cdots + n) + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n(n+1) + 2(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

This completes the induction.

2. A sequence $a_1, a_2, a_3, \ldots$ is *arithmetic* if

$$a_2 - a_1 = a_3 - a_2 = \cdots.$$

Equivalently, there is a number $d \in \mathbb{R}$ such that $a_{n+1} = a_n + d$ for every $n \in \mathbb{N}$. We prove by induction that

$$a_n = a_1 + (n-1)d \qquad (n \in \mathbb{N}).$$

Base case. For $n = 1$, the left side is a_1, and the right side is $a_1 + (1-1)d = a_1$. Thus $P(1)$ is true.

Induction step. Assume $a_n = a_1 + (n-1)d$. We must show $a_{n+1} = a_1 + nd$. Indeed,

$$a_{n+1} = a_n + d = a_1 + (n-1)d + d = a_1 + nd.$$

This completes the proof.

3. Try writing an induction proof for a *geometric* sequence: if $a_{n+1} = qa_n$ for every $n \in \mathbb{N}$, then

$$a_n = a_1 q^{n-1}.$$

Choosing a Suitable Base Case

Example. Let us "prove" that all horses in the world have the same color. More precisely, we will attempt to prove by induction that, for every n, any n horses have the same color.

Base case. For $n = 1$, there is only one horse, and it certainly has the same color as itself.

Induction step. Assume any n horses have the same color. Given $n + 1$ horses, we want to prove they all have the same color.

The first n horses have the same color by the induction hypothesis; call it X. The last n horses, all except the first, also have the same color. The horses in the middle show that this color must be X. Hence all $n + 1$ horses have color X.

The claim is clearly false. Where does the argument fail?

An induction proof need not start at $n = 1$. If the base case or the induction step does not hold there, it is sometimes useful to look for a larger starting value from which both are valid.

Examples.

1. We prove that $2n + 1 < 2^n$ for every integer $n \geq 3$.

   Base case. For $n = 3$,
   $$2n + 1 = 7 < 8 = 2^n.$$

   Induction step. Let $n \geq 3$ and assume $2n + 1 < 2^n$. We must show $2(n + 1) + 1 < 2^{n+1}$. By the induction hypothesis,
   $$\begin{aligned} 2(n+1)+1 &= (2n+1)+2 \\ &< 2^n + 2 \\ &< 2^n + 2^n \\ &= 2 \cdot 2^n = 2^{n+1}. \end{aligned}$$

2. We prove that $n^2 < 2^n$ for every integer $n \geq 5$.

   Base case. For $n = 5$,
   $$n^2 = 25 < 32 = 2^n.$$

   Induction step. Let $n \geq 5$ and assume $n^2 < 2^n$. Using the preceding example,
   $$\begin{aligned} (n+1)^2 &= n^2 + 2n + 1 \\ &< 2^n + (2n+1) \\ &< 2^n + 2^n \\ &= 2 \cdot 2^n = 2^{n+1}. \end{aligned}$$

   Although $n^2 < 2^n$ holds at $n = 1$, that does not make $n = 1$ a valid base for this argument. Otherwise, the result would hold for every natural number, whereas it already fails at $n = 3$. The problem is the induction step: at $n = 1$, the needed estimate $2n + 1 \leq 2^n$ is false.

Summation Notation

In this course, and throughout mathematics, we often encounter sums with a variable number of terms. The summation symbol is convenient in such cases, and in others as well:

$$\sum_{k=1}^{n} a_k = a_1 + \cdots + a_n.$$

Examples.

1. $\sum_{k=1}^{17} 1 = 1 + 1 + \cdots + 1 = 17.$
2. $\sum_{k=1}^{6} k = 1 + 2 + 3 + 4 + 5 + 6 = 21.$
3. $\sum_{k=1}^{n} k = 1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$. We proved this by induction.
4. What is $\sum_{k=1}^{n} (-1)^k$? For example,

$$\sum_{k=1}^{5} (-1)^k = (-1) + 1 + (-1) + 1 + (-1) = -1.$$

Summation symbols can also be nested, something we will encounter later. For example,

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij}.$$

Newton's Binomial Theorem

The identity $(a+b)^2 = a^2 + 2ab + b^2$ is one of the familiar algebraic formulas from school. It is easy to prove:

$$\begin{aligned}(a+b)^2 &= (a+b)(a+b) \\ &= a^2 + ab + ba + b^2 \\ &= a^2 + 2ab + b^2.\end{aligned}$$

There is also a cubic version, likewise easy to prove:

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3.$$

Ignoring the coefficients $1, 2, 1$ and $1, 3, 3, 1$, we see a pattern. The terms are all the products

$$a^0b^2,\ a^1b^1,\ a^2b^0$$

in the first case, or

$$a^0b^3,\ a^1b^2,\ a^2b^1,\ a^3b^0$$

in the second. In general, they are $a^k b^{n-k}$ for $k = 0, \ldots, n$, with $n = 2$ or $n = 3$ here.

There is a general formula for $(a+b)^n$. When $n = 2, 3$, it gives the two identities above. For $n = 4$, for example, it has the form

$$(a+b)^4 = ?\, a^4 + ?\, a^3b + ?\, a^2b^2 + ?\, ab^3 + ?\, b^4.$$

To get a feel for the coefficients, return to $n = 3$ and write

$$(a+b)^3 = (a+b)(a+b)(a+b).$$

After expansion, every summand is a product of three factors, one selected from each pair of brackets. Choosing a, a, a gives a^3. Choosing a, b, a gives a^2b, but so do b, a, a and a, a, b. These are the three possibilities that give a^2b, so the resulting coefficient is 3.

In general,

$$(a+b)^n = \underbrace{(a+b)\cdots(a+b)}_{n \text{ factors}}.$$

There is exactly one way to obtain a^n: choose a from every factor. Thus a^n occurs once.

How many ways are there to obtain $a^{n-1}b$? Choose one of the n factors to contribute b, and choose a from the rest. There are n choices. By the same reasoning,

$$(a+b)^n = a^n + na^{n-1}b + \cdots + nab^{n-1} + b^n.$$

What about the remaining terms? To obtain $a^k b^{n-k}$, choose k of the n factors to contribute a and take b from the others. Thus we ask how many ways there are to choose k objects from a given set of n objects, or equivalently, to form a group of k people from n people.

The answer is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

read "n choose k." As usual, $0! = 1$.

Theorem 5 (Newton's binomial theorem)**.** *For every $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$,*

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

The binomial coefficients have a symmetry,

$$\binom{n}{k} = \binom{n}{n-k}.$$

They have many other interesting properties, but we will not pursue them here.

Examples Using the Binomial Theorem

Examples.

1. Take $n = 4$. The binomial coefficients are

$$\binom{4}{0} = \frac{4!}{0!4!} = 1, \quad \binom{4}{1} = \frac{4!}{1!3!} = 4, \quad \binom{4}{2} = \frac{4!}{2!2!} = 6,$$
$$\binom{4}{3} = \frac{4!}{3!1!} = 4, \quad \binom{4}{4} = \frac{4!}{4!0!} = 1.$$

   Therefore

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

   For example, substituting $a = b = 1$ gives

$$16 = (1+1)^4 = 1 + 4 + 6 + 4 + 1.$$

2. The following is a small trick, but an interesting one:

$$\sum_{k=0}^{n} \binom{n}{k} = \sum_{k=0}^{n} \binom{n}{k} 1^k 1^{n-k} = (1+1)^n = 2^n.$$

2 Fields

2.1 Definition of a Field

Fields

The number systems $\mathbb{Q}$ and $\mathbb{R}$ differ from $\mathbb{N}$ and $\mathbb{Z}$ in that division by every nonzero number is possible. The rational and real numbers are examples of numerical "playgrounds" called fields. Here is a precise definition.

Definition 6 (Field)**.** A *field* is a set $\mathbb{F}$ equipped with two operations, addition and multiplication, satisfying the following conditions.

1. **Binary operations.** For $a, b \in \mathbb{F}$, their sum is denoted by $a + b$ and is also an element of $\mathbb{F}$. The same applies to their product $a \cdot b$.
2. **Associativity of addition and multiplication.** For every $a, b, c \in \mathbb{F}$,
$$a + (b + c) = (a + b) + c,$$
$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$
3. **Commutativity of addition and multiplication.** For every $a, b \in \mathbb{F}$,
$$a + b = b + a, \qquad a \cdot b = b \cdot a.$$
4. **Additive and multiplicative identities.** There are distinct elements $0, 1 \in \mathbb{F}$ such that, for every $a \in \mathbb{F}$,
$$a + 0 = a, \qquad a \cdot 1 = a.$$
5. **Additive inverses.** For every $a \in \mathbb{F}$, there exists $b \in \mathbb{F}$ such that $a + b = 0$.
6. **Multiplicative inverses.** For every nonzero $a \in \mathbb{F}$, there exists $b \in \mathbb{F}$ such that $a \cdot b = 1$.
7. **Distributivity.** For every $a, b, c \in \mathbb{F}$,
$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark. The first condition is sometimes described by saying that $\mathbb{F}$ is closed under addition and multiplication. We will use that terminology in certain later contexts, although it is not quite the right formulation when the operations themselves are part of the definition of a field.

These properties may look obvious, but we will soon see that some fail even in familiar sets of numbers. As the course progresses, we will encounter more sets and operations for which familiar arithmetic properties do not hold.

The operations denoted by $a + b$ and $a \cdot b$ need not be the familiar operations. For example, one can define new operations on integers and ask whether they satisfy the field axioms.

Basic Examples of the Definition

Examples.

1. $\mathbb{N}$ is not a field; for example, it has no additive identity under our convention.
2. $\mathbb{Z}$ is not a field; for example, not every nonzero element has a multiplicative inverse. There is no integer k satisfying $2k = 1$.

   This uses the fact that 1 is the multiplicative identity. Could a different number serve as an identity and make inverses possible? It cannot, and this is easy to check in $\mathbb{Z}$. More generally, we will see that a field cannot have two additive identities or two multiplicative identities.
3. $\mathbb{Q}$, with ordinary addition and multiplication, is a field. Its additive identity is 0, and its multiplicative identity is 1. The additive inverse of m/n is

   $$-\frac{m}{n},$$

   and, when $m \neq 0$, its multiplicative inverse is

   $$\frac{n}{m}.$$

   Of course, all the other conditions in the definition must also be checked.
4. $\mathbb{R}$, with ordinary addition and multiplication, is a field. Since we have not constructed $\mathbb{R}$ precisely, we cannot give a fully rigorous proof of that statement here.

Constructing Fields

Let us try to construct a field with exactly two elements,

$$\mathbb{F} = \{a, b\}.$$

The definition requires two distinct elements, one an additive identity and the other a multiplicative identity. They must therefore be a and b. Suppose a is the additive identity and b is the multiplicative identity.

We prepare addition and multiplication tables that meet the field requirements:

$+$	a	b
a	a	b
b	b	a

$\cdot$	a	b
a	a	a
b	a	b

Think about why each entry must have the displayed value, and check all the field axioms. Try the same construction with $3, 4, 5$, and 6 elements. It may take a while, but the outcomes are interesting.

Basic Properties of a Field

Theorem 7 (Basic field properties)**.** *Let $\mathbb{F}$ be a field. Then:*

1. *The additive identity 0 and the multiplicative identity 1 are unique. For example, if $a, b \in \mathbb{F}$ are both additive identities, then $a = b$.*
2. *Every $a \in \mathbb{F}$ has a unique additive inverse, denoted by $-a$. Every nonzero $a \in \mathbb{F}$ has a unique multiplicative inverse, denoted by a^{-1}.*
3. *For every $a \in \mathbb{F}$, $-(-a) = a$: the additive inverse of the additive inverse is the original element.*
4. *For every nonzero $a \in \mathbb{F}$, $(a^{-1})^{-1} = a$.*
5. *For every $a \in \mathbb{F}$, $a \cdot 0 = 0$.*
6. *If $a, b \in \mathbb{F}$ satisfy $a \cdot b = 0$, then $a = 0$ or $b = 0$.*

In general, nonzero elements whose product is zero are called *zero divisors*. Thus the last property says that a field has no zero divisors.

Before Next Week

It is recommended that you briefly review the following topics before next week:

1. Integer division with remainder and the notion of divisibility, as learned in school.
2. Complex numbers, if you have studied them before.
3. Basic trigonometry: $\sin\theta$ and $\cos\theta$.

> *"Being artificial doesn't mean it isn't real; it only means it has a purpose."*
>
> Brandon Sanderson, *Yumi and the Nightmare Painter*